

Simulation Engine for Marketing ROI and Incremental Value Analysis

1. Summary & Business Purpose

In promotional campaigns, high conversion rates and spikes in gross revenue are not always due to campaigns effectiveness. When financial incentives, such as discounts or free products, are distributed indiscriminately across a customer base, a business risks subsidizing transactions that customers would have completed naturally. In these scenarios, the company absorbs the full cost of the reward while generating zero net-new value.

The primary objective of this project was to architect a robust, data-driven simulation engine capable of evaluating marketing campaigns by measuring their true **incremental value**. By establishing a behavioral counterfactual, calculating how much a customer would have spent in the absence of a promotion, this model isolates the net financial impact of specific offers across distinct customer segments.

The final deliverable of this workflow is an interactive decision-support dashboard. This tool allows to move away from guesswork and to dynamically optimize marketing budget allocation.

Dataset Overview and Dynamics

The foundation of this analysis is the simulated Starbucks Rewards mobile application dataset, a comprehensive portfolio of consumer behavior data documenting how users interact with the brand's promotions. The raw data captures a 30-day chronological window of customer activity and is divided into three JSON files:

- **The Portfolio:** This file acts as the metadata catalog for the marketing department, outlining 10 distinct promotional campaigns. Each campaign is defined by specific operational characteristics, including the `offer_type` (categorized as Buy-One-Get-One, Discount, or Informational), the campaign duration in days, the difficulty (the minimum monetary spend required to trigger the offer), the reward amount

granted to the customer, and the distribution channels utilized, such as email, mobile, social media, and web.

- **The Profile:** This structure contains the demographic information for the customer base. It tracks key user attributes including age, gender, annual income, and the specific date the user created their account (`became_member_on`). Notably, this raw data contains systemic anomalies, such as placeholder profiles explicitly registering an age of 118 with null values for gender and income, which must be accounted for during the analytical modeling phase.
- **The Transcript:** This represents the operational core of the dataset: a highly granular event log capturing hundreds of thousands of customer actions sequentially. It records the exact time (in hours from the start of the campaign) when a specific person executes an event. These events track the full lifecycle of a campaign, logging when an offer is "offer received", "offer viewed", and "offer completed", alongside "transaction" events representing store purchases. Crucially, the granular details of these events, such as the transaction amount or the specific offer ID involved, are housed within a nested value dictionary.

2. Data Preprocessing

The raw Starbucks dataset, consisting of nested JSON objects, is incompatible with business intelligence tools and relational database querying. Before any strategic insights can be extracted, the raw data must be processed, cleaned, and standardized into a flat, relational schema.

Technical Execution: The foundational data engineering and normalization pipeline was constructed utilizing Python and the Pandas library. The objective was to read the raw JSON lines and transform them into structured, analysis-ready tabular formats.

1. Flattening the Portfolio Matrix: Upon ingesting the `portfolio.json` file, a primary structural issue was identified: the marketing channels were stored as a nested list within a single column. To resolve this, I applied vectorized string operations `(str.join('|').str.get_dummies('|'))` to extract the nested list and expand it into individual binary dummy variables. This transformation successfully pivoted the channels into standalone, boolean columns (e.g., separating email, mobile, social, and

web capabilities), which were then concatenated back into the main dataframe while dropping the obsolete original column.

2. Parsing the Transcript Event Logs: The `transcript.json` file presented a more complex challenge. The critical financial and identifying metrics of the entire project—how much money a customer spent, what reward they received, and which specific marketing offer they interacted with, were buried inside a single nested value dictionary column. Furthermore, the data entry within this dictionary was inconsistent, occasionally referencing the offer identifier as `offer_id` and in other instances as `offer_id`.

To systematically extract this data, I deployed targeted Lambda mapping functions across the dataframe:

- I created a new "offer id" column by conditionally extracting either the "offer id" or "offer_id" key from the nested dictionary, ensuring no data was lost due to inconsistencies.
- I extracted the monetary "reward" value into its own dedicated column to track campaign costs.
- I isolated the transaction "amount" into a standalone numerical column, establishing the foundation for all future gross revenue and spending calculations.

3. Relational Standardization and Export: With the nested dictionaries successfully unpacked and the arrays flattened into binary flags, the final step of the preprocessing phase was to enforce a strict relational structure. The obsolete, nested value column was dropped from the transcript dataframe. Finally, the cleaned profile, standardized portfolio, and newly flattened transcript dataframes were exported as CSV files.

3. Metric Engineering & Counterfactual Design

Traditional analytics often automatically attribute a sale to a campaign simply because a discount code was redeemed. However, if a customer receives an offer in their inbox, never opens it, but organically visits a store to make their usual purchase, the system will apply the discount and log a "offer completed." From a financial perspective, this is a total loss: the business just paid a premium to subsidize a transaction that was guaranteed to happen anyway.

To determine the authentic return, one must filter out this "accidental completion" and isolate purely intentional consumer behavior. Furthermore, calculating the actual financial lift generated by a campaign requires establishing a counterfactual baseline, what each specific customer would have spent organically in the total absence of the marketing campaign.

Technical Execution (SQL & DuckDB): To map and evaluate these chronological customer journeys, I engineered a SQL data pipeline utilizing DuckDB and Pandas.

1. Defining Intentional Offer Completions (`is_success`):

Because Starbucks frequently targets the same user with the identical offer over a 30-day period, the events could not be joined simply by user and offer ID.

- I constructed a Common Table Expression (CTE) utilizing Window Functions (`OVER (PARTITION BY person, "offer id" ORDER BY time)`) to dynamically calculate a `campaign_id`, perfectly separating multiple instances of the same offer sent to the exact same customer in different times.
- With unique campaigns isolated, I pivoted the event logs, utilizing conditional aggregations to bring the exact timestamps for `time_received`, `time_viewed`, and `time_completed` onto a single row.
- I then applied a rule using a `CASE WHEN` statement to define true causal success: an offer was marked as intentionally completed (`is_success = 1`) only if the timestamp of the view occurred on or after receipt (`time_viewed >= time_received`), and the completion occurred strictly after the view (`time_completed >= time_viewed`), all before the campaign's expiration deadline (`duration * 24`). Duration represents the time in days a customer has to complete an offer, while the time of the logs is expressed in hours.

2. Isolating the Organic Counterfactual (`is_natural_transaction`):

To calculate the baseline spending habits, I had to separate marketing-influenced purchases from organic ones.

- I executed a `LEFT JOIN` between the standard transactional event log and the newly created `offer_status` table.

- The JOIN condition checked if a transaction timestamp fell within an active, viewed promotional window (`t.time >= s.time_viewed AND t.time <= COALESCE(s.time_completed, s.time_received + (s.duration * 24))`).
- Any transaction that failed to match an active campaign window (`MAX(s."offer_id") IS NULL`) was officially flagged as an `is_natural_transaction=1`.

3. Aggregating User-Level Strategic Metrics:

Finally, the pipeline aggregated these events into two definitive Key Performance Indicators for each individual profile in the database:

- `offer_sensitivity`: Calculated by dividing the sum of intentionally completed offers by the total count of non-informational offers received (`SUM(is_success) / CAST(COUNT(time_received) AS FLOAT)`). This metric precisely quantifies a user's responsiveness to promotions. Informational offers are ignored since they do not neither a cost nor can be completed, consisting of only communications.
- `avg_natural_spending`: Calculated by averaging the monetary amount of purely organic, non-promoted transactions per person. This provides the counterfactual baseline for every user.

4. Customer Segmentation

Treating a diverse customer base as a uniform entity inevitably leads to inefficient budget allocation. While marketing campaigns may increase overall revenue, they may simultaneously destroy profitability by targeting the wrong customers. By evaluating a customer's organic spending against their offer-sensitivity, it's possible to transition from an indiscriminate marketing approach to a proactive, highly targeted budgeting strategy.

Using the two engineered metrics through SQL—*Average Natural Spending* and *Offer Sensitivity*, I segmented the customer base utilizing the median of each metric. This splits the user population into four distinct quadrants:

- **Brand Loyalists**
 - **Profile:** These are the brand champions, comprising approximately 15.38% of the customer base. They exhibit the highest baseline organic spending (21.43 \$ per transaction) and demonstrate the lowest reliance on promotions

(only 22% of all transactions are made when offers are active). Crucially, the data reveals a counterintuitive behavioral shift: when presented with a discount, their average transaction value actually drops to 20.50 \$.

- **Strategic Action:** Cease monetary promotions immediately. The simulation unequivocally proves that sending financial rewards to this cohort results in severe margin cannibalization, generating a negative Incremental Revenue of -2,531.11 \$¹. The business is essentially paying its best customers to spend the same or less.

- **Deal Hunters**

- **Profile:** Making up the largest segment of the population (35.26%), these users are the very profitable for the business. They maintain a strong organic baseline (20.81 \$), but unlike Brand Loyalists, promotions successfully stimulate them to spend even more, pushing their average transaction value up to 22.32 \$ during active campaigns.
- **Strategic Action:** Allocate the majority of the marketing budget here. This segment delivers a 260% ROI and generates the absolute highest positive Incremental Revenue (24,863.52 \$).

- **Occasionals**

- **Profile:** Representing roughly 31.42% of the base, these are low-value buyers. Their natural average spend is already minimal at just 4.77 \$. During promotional windows, their average spend dips even lower to 4.10 \$, despite displaying a high promotional dependence (57%).
- **Strategic Action:** Strictly exclude from all monetary reward campaigns. Generating a -12,815.62 \$ in negative Incremental Revenue and a failing ROAS (98%), promotional investments here explicitly destroy corporate value. Engagement should be strictly limited to zero-cost, purely "Informational" marketing nudges.

- **Promo-Driven**

- **Profile:** This segment (17.94% of users) demonstrates low baseline organic spending (7.30 \$) but is effectively activated by marketing stimuli. When targeted with financial offers, their average transaction value increases to 8.11 \$, maintaining a solid ROI of 150%.

¹ Incremental Revenue is defined as the difference between what a customer would spent organically and what would have spent in presence of promotions. The metric is presented in the next section.

- **Strategic Action:** Deploy targeted upselling campaigns. Because their organic baseline is low, promotional campaigns successfully lift their overall spending floor. This segment yields a Incremental Revenue of 12,192.57 \$.

5. Financial Modeling & Simulation Engine (Excel & DAX)

While understanding historical performance is valuable, predictive tools are required to make forward-looking budget allocation decisions. A theoretical projection that simply multiplies average historical revenues by a proposed marketing budget is fundamentally flawed. It fails to account for real-world operational boundaries, such as the maximum addressable audience size, and ignores market volatility.

To bridge this gap, I engineered a dynamic and constrained simulation engine. It calculates the true incremental lift, the net revenue above the organic counterfactual baseline that was established earlier through SQL. Finally, recognizing that consumer behavior fluctuates, I integrated a bivariate sensitivity matrix to stress-test the campaign's profitability against varying economic conditions.

Technical Execution

The simulation engine was constructed within an Excel sheet, separated from the front-end dashboard. The architecture relies on pulling data directly from the underlying Data Model and DAX metrics.

User Inputs

Budget	10,000 €
Segment	Deal Hunters
Offer type	discount
Stress Factor Conv	0%
Stress Factor Avere	0%

Results

Simulated Conv. Ra	70.30%
Expected offers	3,462
Expected Conversic	2,433
Expected Revenue	54,311.26 €
Expected Cost	9,999.94 €
Expected Profit	44,311.32 €
Budget Left	0.06 €
Expected ROI	443%
Simulated promo sp	22.32 €
Simulated natural s	20.81 €
Incremental Revenu	3,679.40 €

Calculations

Segments	Target Users	Offer Type
Brand Loyalists	5,000	bogo
Deal Hunters	10,000	discount
Promo Driven	15,000	
Occasionals	20,000	
Sendable offers		
Financial limit	3462	
Physical limit	4986	

Metrics

Conversion Rate	70.30%
Average promo spe	22.32 €
Average reward	2.89 €
Average natural spe	20.81 €
ROAS	543%
Total transactions	16,441
Stress factor (conv.	20
Population	4986
Stress factor (spent	20

Waterfall Chart

Category	Value
Natural Spent	50,631.86 €
Incremental Revenue	3,679.40 €
Promo Spent	54,311.26 €
Cost	9,999.94 €
Profit	44,311.32 €

1. Dynamic Data Extraction via CUBEVALUE. To build a resilient and responsive engine, I avoided using lookup functions on pivot tables and directly queried the DAX Data Model using `CUBEVALUE` functions.

- In the “Metrics” block, the engine extracts the core historical averages tailored to the specific selections made by the user on the dashboard.
- Formulas such as `=CUBEVALUE("ThisWorkbookDataModel", ...)` dynamically retrieve the segment's Conversion Rate, Average promo spent, Average reward cost, Average natural spent (the counterfactual baseline), ROI, and the total available Population for the selected segment.
- This ensures that every time a stakeholder changes a dropdown menu on the executive dashboard, the fundamental building blocks of the simulation update instantaneously.

2. Implementing Market Volatility (Stress Factors) Marketing performance is rarely static. To allow to model optimistic or pessimistic scenarios, I integrated two "Stress Factors" linked to dashboard spin buttons.

- In the “Metrics” section, baseline stress values are captured (e.g., Stress factor (conv.) = 20).
- In the “User Inputs” block, these raw integers are converted into functional percentage. For example, the Stress Factor Conversion is calculated as $\text{Stress Factor} - 20/100$, transforming a base input of 20 into a 0% baseline, or an input of 15 into a -5% pessimistic adjustment. 20 is subtracted as Excel does not allow negative number for the spin button. In this way, the stress factor can vary between -20% and +20%.
- These modifiers directly feed into the “Results” section to create simulated scenarios: Simulated Conv. Rate is calculated as $\text{Conversion Rate} \times (1 + \text{Converted Stress Factor})$ adjusting the historical conversion rate up or down based on the risk hypothesis. Similar logic is applied to generate the Simulated promo spent and Simulated natural spent.

3. The Operational Funnel and Capacity Constraints A campaign cannot spend more money than it is allocated, nor can it send offers to more people than exist in the segment.

I built strict mathematical constraints in the **Calculations** block to enforce these boundaries:ia

- **The Financial Limit:** The engine calculates the maximum number of rewards the budget can support via Total Budget divided by the Average Reward Cost. The function ensures the business only models complete
- **The Physical Limit:** This is strictly bound to the target Population extracted via the CUBE formulas.
- **Sendable Offers:** In the **Results** column, the actual Expected offers deployed in the simulation is determined by the minimum between the two limits, forcing the engine to select whichever constraint (budget or audience size) is hit first. An adjacent formula, automatically flags “Max” if the campaign has saturated the entire physical market.

4. Executing the Incremental Profit Logic With the operational boundaries set and the behavior simulated, the engine computes the final financial impact in the “Results” block.

- **Volume:** Expected Conversions are calculated by multiplying the sendable offers by the simulated conversion rate.
- **The Vanity Metrics:** Expected Revenue evaluates the gross intake, *and* Expected Cost computes the total reward expenditure.
- **The True Value (Counterfactual):** The most critical calculation in the entire engine is the *Incremental Revenue*. Instead of looking at gross sales, the formula isolates the financial lift. It multiplies the total conversions by the difference between what the customer would have spent during the promotion (Simulated promo spent) and would have spent in the absence of it (Simulated natural spent). There are based on the average observed spending for each segment. If the incremental revenue is positive, then the campaign would effectively stimulate the segment in spending more than usual.
- **The Bottom Line:** The final Expected Profit ($=F5-F6$) and Expected ROAS ($=(F5-F6)/F6$) definitively prove whether the campaign generated net-new wealth for the company or simply burned capital.

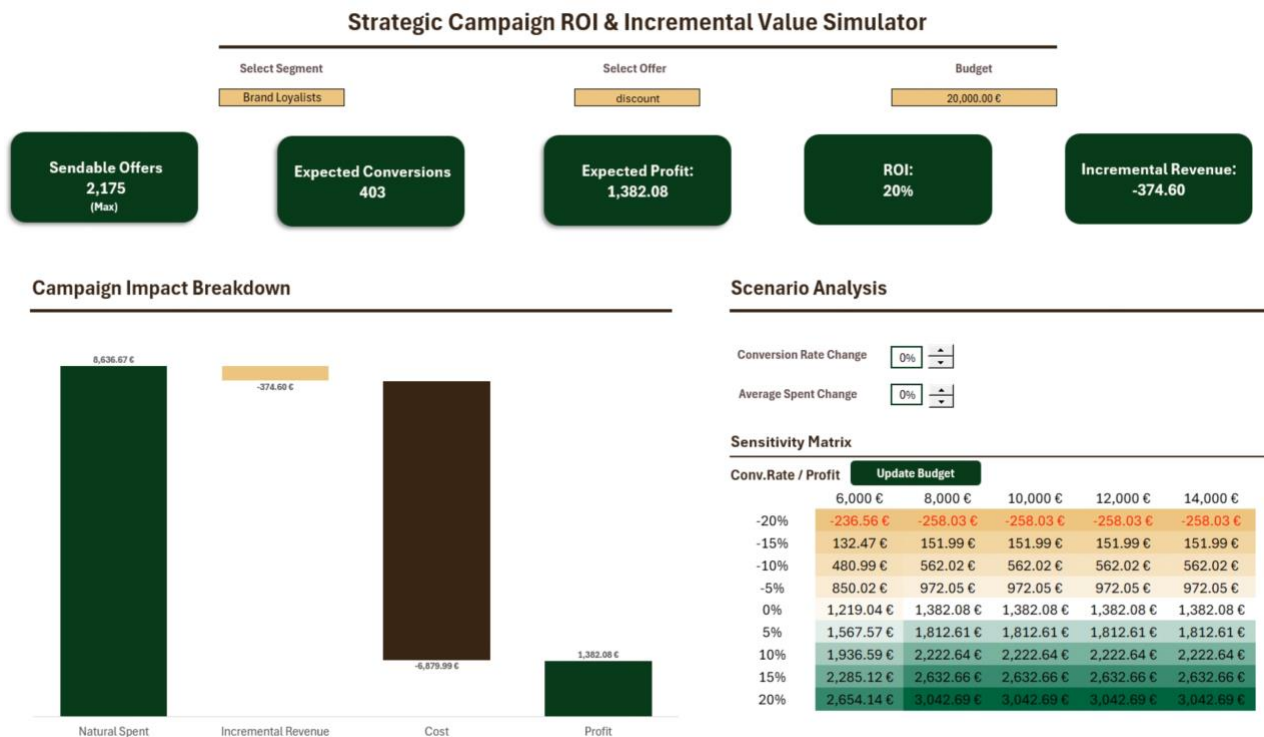
Dynamic data table				
12,000 €	24,000 €	30,000 €	36,000 €	42,000 €

Conv.Rate / Profit						
		4,000 €	8,000 €	10,000 €	12,000 €	14,000 €
-20%		13,369.44 €	26,758.31 €	33,462.46 €	40,147.18 €	46,831.89 €
-15%		14,440.93 €	28,923.61 €	36,163.51 €	43,406.30 €	50,626.76 €
-10%		15,534.74 €	31,088.92 €	38,886.89 €	46,665.42 €	54,443.95 €
-5%		16,628.56 €	33,276.55 €	41,610.26 €	49,924.54 €	58,238.82 €
0%		17,700.05 €	35,441.86 €	44,311.32 €	53,183.66 €	62,033.69 €
5%		18,793.86 €	37,607.16 €	47,034.69 €	56,442.79 €	65,850.88 €
10%		19,887.68 €	39,794.79 €	49,758.07 €	59,701.91 €	69,645.75 €
15%		20,959.17 €	41,960.10 €	52,459.12 €	62,961.03 €	73,440.62 €
20%		22,052.99 €	44,125.41 €	55,182.50 €	66,220.15 €	77,257.81 €

5. Waterfall chart and The Sensitivity Matrix To translate these raw calculations into visualizations, I structured two final components.

- **Waterfall Chart Mapping:** I built a dedicated table to feed the dashboard's visual value bridge. It anchors the organic Simulated Natural Spent, maps the positive or negative Incremental Revenue, tracks the total Simulated Promo Spent and Cost, and outputs the final Profit. This allows to visualize the process through which a customer segment, starting with their baseline spending behavior, increase or decrease their expenditure due to the campaign (incremental revenue), and how this translates into the final profit after accounting for the cost.
- **Two-Variable Data Table (Heatmap):** To provide immediate risk assessment, I constructed a dynamic sensitivity table at the bottom of the calculation sheet. The row headers represents budget variations ranging between -40% and + 40%. The column headers represent conversion rate stress shifts from -20% to +20%. The matrix computes the expected profit for each combination. Conditional formatting allows to instantly visually identify the maximum risk exposure before committing a single dollar of actual marketing budget. The data table functionality allows only hard-coded values for the budget. The line below “Dynamic data table” takes the selected budget from the dashboard and computes the variation. A script will past those values into the row headers through a button in the dashboard.

6. Executive Dashboard



The objective of this final phase was to construct a clean executive interface, designed to visually expose the core business problem this project set out to solve. The dashboard forces decision-makers to evaluate campaigns based on true incremental value rather than metrics like gross conversion volume.

Technical Execution

The dashboard is structured into three interactive, modular zones, each serving a distinct strategic purpose:

1. The Strategic Control Center and KPI Banner

The top tier of the dashboard acts as the primary user interface, featuring clean dropdown controls and dynamic summary cards.

- **User Inputs:** Stakeholders can define their simulation parameters by selecting a target demographic from the Select Segment dropdown (e.g., "Brand Loyalists"), choosing an offer type from the Select Offer menu (e.g., "discount"), and allocating a maximum financial constraint via the Budget input (e.g., 30,000.00 \$).

- **Dynamic KPIs:** Instantly upon selection, the underlying engine recalculates and feeds five primary KPI cards.
- **The Campaign Real Value:** The displayed KPIs perfectly illustrate the exact business problem this project solves. In the specific scenario simulated on the dashboard (targeting Brand Loyalists with a discount), the campaign successfully generates 403 Conversions and an Expected Profit of 1382.08 \$, yielding a seemingly positive ROAS of 20%.. However, the final KPI card reveals the harsh truth: the Incremental Revenue is -374.60 \$. The dashboard proves that the business just spent thousands of euros subsidizing loyal customers who would have spent more money organically without the discount.

2. The Value Bridge (Campaign Impact Breakdown)

To ensure stakeholders fundamentally understand how that negative incremental revenue was calculated, I implemented a Waterfall chart.

- This visual narrative anchors the exact counterfactual baseline, the Natural Spent as the starting pillar on the left.
- It then maps the negative dip of the Incremental Revenue
- Finally, it visually subtracts the marketing Cost block to arrive at the Profit
- By visualizing the journey from the counterfactual to the final profit, the chart forces the user to see that the profit generated was entirely cannibalized from the natural baseline, destroying organic value.

3. Scenario Analysis & The Risk Sensitivity Matrix

Because consumer behavior is highly susceptible to market volatility, the final module allows to stress-test their campaign allocations against varying economic realities.

- **Macro-Economic Stress Testing:** I integrated Excel Form Controls (spin buttons) allowing users to adjust the Conversion Rate Change and Average Spent Change in granular percentage increments.
- **The Heatmap Matrix:** Below the controls lies the true decision-support engine: a two-variable Sensitivity Matrix (Conv.Rate / Profit). The vertical axis maps the simulated conversion rate shifts (from -20% to +20%), while the horizontal axis scales variations of the user's base budget (ranging from 1,200 \$ to 4,200 \$). The

button “Update budget” allows to quickly refresh the budgets in the row headers whenever it is changed in the first dedicated cell.

- **Automated Risk Identification:** Through the Data Table, the matrix computes the expected profit for every possible intersection of budget and conversion rate. I applied conditional formatting to create a heatmap.

This final module transforms the project from a static reporting tool into a risk-mitigation platform.